BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective Type) - Group - I

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The half life of a certain isotope is one day. What is the quantity of the isotope after two days :

(A) One-half
(B) One-quarter
(C) One-eighth
(D) One-sixteenth

Q2. The brain of any computer system is :

(A) Monitor
(B) Memory
(C) CPU
(D) Control unit

Q3. The particles emitted from a hot cathode surface are called :

(A) Positive ions
(B) Negative ions
(C) Protons
(D) Electrons

Q4. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire :

(A) Increases
(B) Decreases
(C) Remains the same
(D) Will be zero

Q5. What is the power rating of a lamp connected to a 12V source when it carries 2.5A current:

(A) 60 W
(B) 30 W
(C) 14.5 W
(D) 4.8 W

Q6. Capacitance is defined as :

(A) VC
(B) QV
(C) Q/V
(D) V/Q

Q7. The index of refraction depends on :

(A) The object distance
(B) The image distance
(C) The focal length
(D) The speed of light

Q8. How does sound travel from its source to your ear:

(A) By change in air pressure
(B) By vibration in wire or string
(C) By electromagnetic waves
(D) By infrared waves

Q9. When water waves enters the region of shallow water their wave length :

(A) Remain same
(B) Become zero
(C) Increases
(D) Decreases

Q10. Intensity of sound of lawn mower:

(A) $10^{1}Wm^{-2}$
(B) $10^{-2}Wm^{-2}$
(C) $10^{-5}Wm^{-2}$
(D) $10^{-8}Wm^{-2}$

Q11. Half life of Cobalt $^{60}_{27}Co$ is :

(A) 2.85 years
(B) 12.3 years
(C) 30 years
(D) 40 years

Q12. Power of hair dryer is :

(A) 1500 watts
(B) 1200 watts
(C) 800 watts
(D) 1000 watts

Physics (Essay Type) (PART - I) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) What are two components of CRO?

(ii) Draw a labelled diagram of Ball and Bowl System.

(iii) State the laws of refraction of light.

(iv) What is restoring force? Give an example.

(v) Define electromagnetic waves, with an example.

(vi) What is electron gun?

(vii) Differentiate between convex and concave mirrors.

(viii) What is meant by critical angle? Draw its diagram.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Define loudness of sound.

(ii) What is meant by intensity level of sound? What is its SI unit?

(iii) Define capacitor and write its formula.

(iv) Draw the label diagram of electroscope.

(v) Which instrument is used to observe sound waves?

(vi) What is meant by primary memory?

(vii) What is ATM? Also write its any one advantage.

(viii) What is Electronic Mail (E-Mail)? Also write its one benefit.

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) What is mutual induction?

(ii) Describe Fleming's left hand rule to find the direction of force.

(iii) Differentiate between direct current and alternating current.

(iv) Differentiate between natural and artificial radioactivity.

(v) What is cosmic radiations? Give examples.

(vi) How potential difference can be measured?

(vii) How may watt hours in 1000 joules?

(viii) What is ionization? What is its power?

Physics (Essay Type) (PART-II) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 4) What is prism? Explain with diagram the path of light through the prism. Also define angle of deviation.

Q5(b). (Marks 5) A pendulum of length 0.99m is taken to the moon by an astronaut. The period of the pendulum is 4.9S. What is the value of 'g' on the surface of the moon?

Q6(a). (Marks 4) Explain the electrostatic by example of electrostatic powder spray painting.

Q6(b). (Marks 5) If at Anarkali Bazaar Lahore intensity level of sound is 80 dB, what will be the intensity of sound there?

Q7(a). (Marks 4) What is meant by Half Life? Explain with the help of graph.

Q7(b). (Marks 5) A step-up transformer has a turn ration 1 : 100. An alternating supply of 20V is connected across the primary coil. What is the secondary voltage?

GROUP – II

Physics (Objective Type) - Group - II

Time Allowed: 15 Min. | **Maximum Marks:** 12

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which of the following is not processing:

(A) Arranging
(B) Manipulating
(C) Calculating
(D) Gathering

Q2. A device which convert mechanical energy into electrical energy is called:

(A) A. C generator
(B) D.C. motor
(C) Transformer
(D) Electroscope

Q3. Speed of light in water is:

(A) $2.3 \times 10^8\ ms^{-1}$
(B) $3.0 \times 10^8\ ms^{-1}$
(C) $2.5 \times 10^8\ ms^{-1}$
(D) $2.0 \times 10^8\ ms^{-1}$

Q4. The process by which electrons are emitted by a hot metal surface is known as:

(A) Boiling
(B) Evaporation
(C) Conduction
(D) Thermionic emission

Q5. Which part of D.C motor reverses the direction of current through the coil every half cycle:

(A) Armature
(B) Commutator
(C) Brushes
(D) Slip rings

Q6. The loudness of a sound is most closely related to its:

(A) Frequency
(B) Period
(C) Wavelength
(D) Amplitude

Q7. Image formed by a camera is:

(A) Real, inverted and diminished
(B) Virtual, upright and diminished
(C) Virtual, upright and magnified
(D) Real, inverted and magnified

Q8. Capacitance is defined as:

(A) VC
(B) Q/V
(C) QV
(D) V/Q

Q9. The number of neutrons in a Uranium $^{238}_{92}U$ isotope are:

(A) 92
(B) 146
(C) 238
(D) 330

Q10. When did Graham Bell make a simple telephone:

(A) 1876
(B) 1886
(C) 1870
(D) 1867

Q11. What happens to the intensity or the brightness of the lamps connected in series as more and more lamps are added:

(A) Increases
(B) Decreases
(C) Remains the same
(D) Cannot be predicted

Q12. When a heavy nucleus splits into two lighter nuclei, the process would be:

(A) Release nuclear energy
(B) Absorb nuclear energy
(C) Release chemical energy
(D) Absorb chemical energy

Physics (Essay Type) (PART - I) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Write the two important features of simple harmonic motion.

(ii) What is meant by damped oscillations?

(iii) What is restoring force? Give an example.

(iv) Show the types of reflection with diagram.

(v) Define Snell's law. Write its formula.

(vi) What is meant by digitization?

(vii) Write the symbol and truth table for NOT gate.

(viii) Explain the deflection of electrons by electric field.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Define quality of sound.

(ii) Calculate the sound intensity level of rustling of leaves.

(iii) Define electric field intensity.

(iv) In what direction will a positively charged particle move in an electric field?

(v) Draw the electric field lines for two oppositely and equal point charges.

(vi) Write name of the components of computer based information system.

(vii) Why optical fibre is more useful tool for the communication process?

(viii) What is the difference between compact disk and floppy disk?

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) Define natural radioactivity.

(ii) Write working principle of electric motor.

(iii) What is difference between ohmic and non-ohmic materials?

(iv) State Joule's law. Write its equation.

(v) What is Cosmic radiations? Give examples.

(vi) Can a transformer operate on D.C.?

(vii) What is Lenz's law?

(viii) Write two isotopes of hydrogen.

Physics (Essay Type) (PART-II) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 4) Explain the two types of mechanical waves with diagram.

Q5(b). (Marks 5) An object 10 cm in front of convex mirror forms an image 5 cm behind the mirror. What is the focal length of the mirror?

Q6(a). (Marks 4) What is flow of information? Explain it with diagram.

Q6(b). (Marks 5) The force of repulsion between two identical positive charges is 0.8N, when the charges are 0.1m apart. Find the value of each charge.

Q7(a). (Marks 4) Define an electromagnet. Make the relay switch circuit diagram and discuss the electromagnet role in this case.

Q7(b). (Marks 5) Cobalt-60 is a radioactive element with half life of 5[illegible in source scan] years. What fraction of the original sample will be left after 26 years.